

# FedMAQ: Multi-Adaptive Quantization & Knowledge Distillation

Communication-Efficient Federated Learning

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June 23, 2026

- We shifted focus from physical IoT/Smart Campus energy forecasting to classical Federated Learning.
- We chose image classification (e.g., MNIST, CIFAR) to make experiments more manageable and reproducibility easier.
- We want to explore multi-adaptive quantization with server-side knowledge distillation with the hopes of further reducing communication while maintaining comparable accuracy.

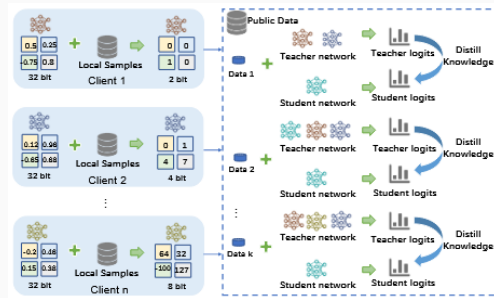
## (Some) Related Works on Communication-Efficient FL

| Algorithm            | Time-Aware | Data-Aware | Resource-Aware | Uses KD    | Key Components                         |
|----------------------|------------|------------|----------------|------------|----------------------------------------|
| DynFed               | Yes        | Yes        | No             | Yes        | Quantization-Aware KD                  |
| AdaDQ-KD             | No         | No         | Yes            | Yes        | Dithering Quantization & Diff. Privacy |
| DAdaQuant            | Yes        | Yes        | No             | No         | Doubly-Adaptive Fixed-Point            |
| FedDT                | No         | No         | No             | Yes        | Ternary Compression                    |
| LAQ-HC               | No         | Yes        | Yes            | No         | Reinforcement Learning                 |
| FedPAQ               | No         | No         | No             | No         | Periodic Averaging                     |
| <b>FedMAQ (Ours)</b> | <b>Yes</b> | <b>Yes</b> | <b>Yes</b>     | <b>Yes</b> | <b>3D-Awareness + Logit KD</b>         |

We explore simultaneously considering client resource constraints, data skew, and temporal training dynamics.

# FedMAQ Architecture

- The server broadcasts the global model parameters to participating client nodes.
- Clients perform local optimization steps on their non-IID datasets.
- Uplinks are compressed using multi-adaptive quantization based on live bandwidth and stability.
- The server aggregates client logits via knowledge distillation to neutralize quantization noise.

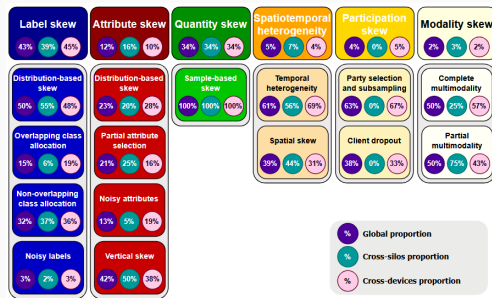


He et al. (2025 - DynFed), Figure 2: Overview of the training procedure using logits.

Our work is a logical extension of DynFed by He et al (2025).

# Experimental Setup

- Five standard vision benchmarks are selected: MNIST, FMNIST, CIFAR-10, CIFAR-100, and FEMNIST.
- Data distributions across clients are skewed using a Dirichlet concentration parameter  $\alpha \in \{0.1, 1, 10\}$ .
- Bandwidth bottlenecks are simulated using a uniform distribution  $U(5, 20)$  Mbps.



Jimenez et al. (2024), Figure 7: Bubble chart of class distributions under Dirichlet skew.

Statistical and system heterogeneity are modeled simultaneously to reflect realistic edge environments.

# Evaluation Stack

## **Foundational Baselines**

FedAvg, FedProx

## **Static Compression**

DAdaQuant, FedPAQ

## **Pure Distillation**

FedKD, FedDistill

## **Adaptive Compression**

AdaGQ, LAQ-HC

## **Direct SOTA Competitors**

DynFed, AdaDQ-KD, FedDT

An 11-algorithm evaluation stack benchmarks FedMAQ against compression, distillation, and adaptive SOTA.

| Chapter                                 | Status & Accomplishments                                                     |
|-----------------------------------------|------------------------------------------------------------------------------|
| <b>Chapter 1: Introduction</b>          | Outlined (RQ, objectives, scope of 5 datasets).                              |
| <b>Chapter 2: Related Works</b>         | Outlined (SOTA algorithm comparisons and literature gaps).                   |
| <b>Chapter 3: Theoretical Framework</b> | Outlined (Mathematical foundations for gradient variance and KD divergence). |
| <b>Chapter 4: Methodology</b>           | Outlined (Data collection, the 4-phase training loop, evaluation metrics).   |

## Next Week Commitments

- Configure the Flower-based Federated Learning simulation environment for Phase 1 baselines.
- Implement Dirichlet non-IID partitioning pipelines across the selected vision datasets.
- Draft the introduction and literature review chapters of the thesis proposal manuscript.

**Thank You!**

**Questions & Answers**